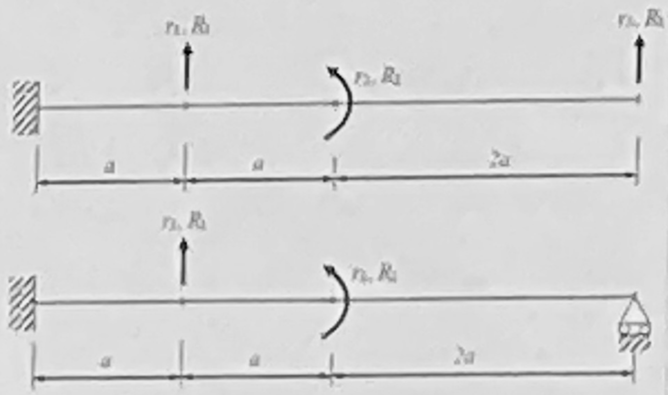
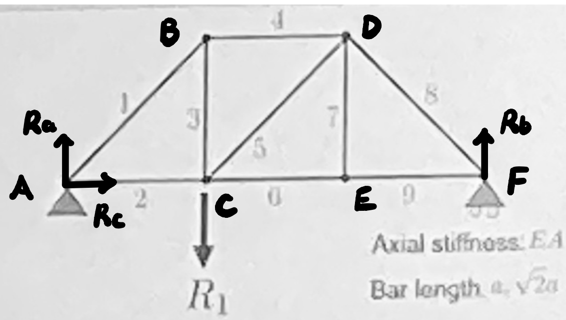
Problem to solve:



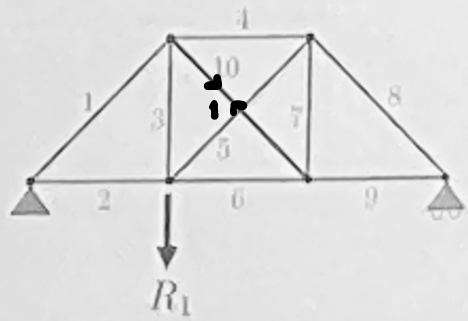
- Given:

$$F = \frac{a}{6EI}\begin{bmatrix} 2a^2 & 3a & 11a^2 \\ 3a & 12 & 36a \\ 11a^2 & 36a & 128a^2 \end{bmatrix}$$

Find the new 2 × 2 flexibility matrix for the structure.



- Determine the axial force acting on each bar.



- Determine the force in the extra bar (bar 10)

1. For the statically indetermine structure:

$$\begin{bmatrix} F_{11} & F_{12} & F_{13} \\ F_{21} & F_{22} & F_{23} \\ F_{31} & F_{32} & F_{33} \end{bmatrix}\begin{bmatrix} R_1 \\ R_2 \\ R_3 \end{bmatrix} = \begin{bmatrix} r_1 \\ r_2 \\ 0 \end{bmatrix}$$

$$\therefore \begin{bmatrix} F_{11} & F_{12} \\ F_{21} & F_{22} \end{bmatrix}\begin{bmatrix} R_1 \\ R_2 \end{bmatrix} + \begin{bmatrix} F_{13} \\ F_{23} \end{bmatrix} R_3 = \begin{bmatrix} r_1 \\ r_2 \end{bmatrix}$$

$$\text{and } \begin{bmatrix} F_{31} & F_{32} \end{bmatrix}\begin{bmatrix} R_1 \\ R_2 \end{bmatrix} + F_{33} R_3 = 0$$

$$\therefore R_3 = -(F_{33})^{-1}\begin{bmatrix} F_{31} & F_{32} \end{bmatrix}\begin{bmatrix} R_1 \\ R_2 \end{bmatrix}$$

$$\therefore \begin{bmatrix} F_{11} & F_{12} \\ F_{21} & F_{22} \end{bmatrix}\begin{bmatrix} R_1 \\ R_2 \end{bmatrix} - \begin{bmatrix} F_{13} \\ F_{23} \end{bmatrix}(F_{33})^{-1}\begin{bmatrix} F_{31} & F_{32} \end{bmatrix}\begin{bmatrix} R_1 \\ R_2 \end{bmatrix} = \begin{bmatrix} r_1 \\ r_2 \end{bmatrix}$$

$$\therefore \begin{bmatrix} G_{11} & G_{12} \\ G_{13} & G_{14} \end{bmatrix} = \begin{bmatrix} F_{11} & F_{12} \\ F_{21} & F_{22} \end{bmatrix} - \begin{bmatrix} F_{13} \\ F_{23} \end{bmatrix}(F_{33})^{-1}\begin{bmatrix} F_{31} & F_{32} \end{bmatrix}$$

$$= \frac{a}{6EI}\left(\begin{bmatrix} 2a^2 & 3a \\ 3a & 12 \end{bmatrix} - \begin{bmatrix} 11a^2 \\ 36a \end{bmatrix}\frac{1}{128a^2}\begin{bmatrix} 11a^2 & 36a \end{bmatrix}\right)$$

$$= \frac{a}{6EI}\left(\begin{bmatrix} 2a^2 & 3a \\ 3a & 12 \end{bmatrix} - \begin{bmatrix} \frac{121}{128}a^2 & \frac{99}{32}a \\ \frac{99}{32}a & \frac{81}{8} \end{bmatrix}\right)$$

$$= \frac{a}{6EI}\begin{bmatrix} \frac{135}{128}a^2 & -\frac{3}{32}a \\ -\frac{3}{32}a & \frac{15}{8} \end{bmatrix}$$

$$= \frac{a}{768EI}\begin{bmatrix} 135a^2 & -12a \\ -12a & 240 \end{bmatrix}$$

2. $\Sigma F_h = R_c = 0$

$\Sigma F_v = R_a + R_b = R_1$

$\Sigma M(A) = R_1 a - 3R_b a = 0 \quad \therefore R_b = \frac{1}{3}R_1, \ R_a = \frac{2}{3}R_1$

$\therefore$ At A, $\Sigma F_v = \frac{\sqrt{2}}{2}N_1 + R_a = 0 \quad \therefore N_1 = -\frac{2}{3}\sqrt{2}R_1$

$\Sigma F_h = \frac{\sqrt{2}}{2}N_1 + N_2 = 0 \quad \therefore N_2 = \frac{2}{3}R_1$

At B, $\sum F_v = N_3 + \frac{\sqrt{2}}{2}N_1 = 0$ $\therefore N_3 = \frac{2}{3}R_1$

$\sum F_h = N_4 - \frac{\sqrt{2}}{2}N_1 = 0$ $\therefore N_4 = -\frac{2}{3}R_1$

At C, $\sum F_v = N_3 + \frac{\sqrt{2}}{2}N_5 - R_1 = 0$ $\therefore N_5 = \frac{\sqrt{2}}{3}R_1$

$\sum F_h = N_2 - \frac{\sqrt{2}}{2}N_5 - N_6 = 0$ $\therefore N_6 = \frac{1}{3}R_1$

At E, $\sum F_v = N_7 = 0$ $\therefore N_7 = 0$

At F, $\sum F_v = \frac{\sqrt{2}}{2}N_8 + R_b = 0$ $\therefore N_8 = -\frac{\sqrt{2}}{3}R_1$

$\sum F_h = N_9 + \frac{\sqrt{2}}{2}N_8 = 0$ $\therefore N_9 = \frac{1}{3}R_1$

When add bar 10, $\bar{N}_{10} = 1$

$\sum F_h = R_c = 0$

$\sum F_v = R_a + R_b = 0$

$\sum M(A) = -3R_b = 0$ $\therefore R_b = 0$ $\therefore R_a = 0$

$\therefore$ At A, $N_1 = 0$, $N_2 = 0$

At B, $\sum F_v = \frac{\sqrt{2}}{2}N_1 + N_3 + \frac{\sqrt{2}}{2}N_{10} = 0$ $\therefore N_3 = -\frac{\sqrt{2}}{2}$

$\sum F_h = N_4 + \frac{\sqrt{2}}{2}N_{10} - \frac{\sqrt{2}}{2}N_1 = 0$ $\therefore N_4 = -\frac{\sqrt{2}}{2}$

At C, $\sum F_v = N_3 + \frac{\sqrt{2}}{2}N_5 = 0$ $\therefore N_5 = 1$

$$\Sigma F_h = \frac{\sqrt{2}}{2}N_5 + N_6 - N_2 = 0 \qquad \therefore N_6 = -\frac{\sqrt{2}}{2}$$

$$\text{At } E,\ \Sigma F_v = N_7 + \frac{\sqrt{2}}{2}N_{10} = 0 \qquad \therefore N_7 = -\frac{\sqrt{2}}{2}$$

$$\text{At } F,\ N_8 = 0,\ N_9 = 0$$

$$\therefore \delta_{10} = \Sigma \frac{N_b^{R_1}\bar{N}_b^{10}}{EA}L_b$$

$$= (0 + 0 - \frac{1}{3}a + \frac{1}{3}a + \frac{2}{3}a - \frac{\sqrt{2}}{6}a + 0 + 0 + 0 + 0)R_1$$

$$= (\frac{4-\sqrt{2}}{6}a)R_1$$

$$\bar{\delta}_{10} = \Sigma \frac{N_b^{10}\bar{N}_b^{10}}{EA}L_b$$

$$= 0 + 0 + \frac{1}{2}a + \frac{1}{2}a + \sqrt{2}a + \frac{1}{2}a + \frac{1}{2}a + 0 + 0 + \sqrt{2}a$$

$$= (2 + 2\sqrt{2})a$$

$$\therefore \delta_{10} + X\bar{\delta}_{10} = 0$$

$$\therefore N_{10} = X = -\frac{\delta_{10}}{\bar{\delta}_{10}} = \frac{-6+5\sqrt{2}}{12}R_1 \qquad \frac{6-5\sqrt{2}}{12}R_1$$